

Problem 1: Derivation of the Logistic Regression Gradient

i) $\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$, $z = x^T w + b$

$$f_{CE}(w, b) = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

Show that $\nabla_w f_{CE}(w, b) = -x(y - \hat{y})$

$$\hat{y}(w, b) = \frac{1}{1 + e^{-(wx+b)}} = (1 + e^{-wx-b})^{-1}$$

$$\frac{\partial \hat{y}}{\partial w} = -(1 + e^{-wx-b})^{-2} \times (-x e^{-wx-b}) = \frac{x e^{-wx-b}}{(1 + e^{-wx-b})^2}$$

$$= \frac{x}{(1 + e^{-wx-b})} \times \frac{e^{-wx-b}}{1 + e^{-wx-b}} = \frac{x}{1 + e^{-wx-b}} \times \frac{1 + e^{-wx-b} - 1}{1 + e^{-wx-b}}$$

$$= \frac{x}{1 + e^{-wx-b}} \left(\frac{1 + e^{-wx-b}}{1 + e^{-wx-b}} - \frac{1}{1 + e^{-wx-b}} \right)$$

$$\nabla_w \hat{y} = x \hat{y} (1 - \hat{y})$$

$$\nabla_{\hat{y}} f_{CE}(w, b) = \frac{\partial}{\partial \hat{y}} (-y \log \hat{y}) + \frac{\partial}{\partial \hat{y}} ((1-y) \log(1-\hat{y})) = -$$

$$= \frac{-y}{\hat{y}} + \frac{(1-y)}{1-\hat{y}} = \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{\hat{y}(1-\hat{y})} = \frac{-y + \hat{y}y + \hat{y} - \hat{y}y}{\hat{y}(1-\hat{y})}$$

$$= \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

$$\nabla_w f(w, b) = \frac{\partial f}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \times x \hat{y} (1-\hat{y}) = \boxed{x(\hat{y} - y) = -x(y - \hat{y})}$$

ii) $\nabla_b \hat{y} = \hat{y}(1-\hat{y})$

$$\Rightarrow \nabla_b f(w, b) = \frac{\partial f}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \times \hat{y}(1-\hat{y}) = \hat{y} - y = \boxed{-(y - \hat{y})}$$

iii) Generalizing the average loss across a mini batch of n observations

$$\nabla_{\omega} f_{CE}(\omega, b) = \frac{1}{n} \sum_{i=1}^n x^{(i)} (y^{(i)} - \hat{y}^{(i)})$$

$$\nabla_b f_{CE}(\omega, b) = \frac{1}{n} \sum_{i=1}^n y^{(i)} - \hat{y}^{(i)}$$